

6.2.a Geophysical Well Log Report

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Executive Summary

Phase II of *Project ECO₂S* evaluated the viability of deep underground CO₂ storage in Kemper County, Mississippi as a part of the Dept. of Energy (DOE) and National Energy Technology Laboratory (NETL) CarbonSAFE initiative. This report synthesizes the various forms of geological well log data that were compiled and gathered from subsurface wells that penetrate the storage geology at or in the vicinity of the CO₂ storage complex in Kemper County. Both routine and non-routine geophysical well logs in addition to limited core were used to evaluate several stacked saline reservoirs and confining units from a combination of legacy oil/gas wells, a deep water production test well, and three *Project ECO₂S* wells drilled in 2017.

Routine geophysical well logs include industry standard logging tools such as the differential caliper, spontaneous potential, gamma ray, neutron porosity, and density porosity among other forms of subsurface data. These well logs were typically available due to the presence of older oil and gas wells that occurred in and around Kemper County. In addition, three wells (MPC 26-5 #1, MPC 34-1 #1, MPC 10-4 #1) were drilled in 2017 for the *ECO₂S* Project and provided new Triple Combo logs and mud logs of the storage geology immediately within the project site.

Non-routine processed geophysical well logs include well logging instruments that measure advanced petrophysical properties from the target formation. Non-routine geophysical logs utilized at the new *ECO₂S* wells include Combined Magnetic Resonance (CMR), Dipole Sonic, and Formation Micro-Imaging (FMI). Using a combination of core, density/neutron porosity, and CMR, the porosity and permeability of the saline storage reservoirs were estimated. Importantly, these determinations of reservoir quality were used to estimate the CO₂ storage capacity at the *ECO₂S* Project site and to model the reservoir performance during CO₂ injection.

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1.0 Introduction

The Phase II *Project ECO₂S* is a DOE/NETL CarbonSAFE initiative to investigate the viability of deep underground CO₂ injection and storage in Kemper County, Mississippi, **Figure 1**. The overarching goal of this demonstration includes four phases (I – IV) that aim to construct a fully characterized and permitted regional CO₂ Storage Complex able to accept commercial scale volumes of CO₂. The Kemper Regional Storage Complex captures CO₂ from Mississippi Power Company's Kemper County energy facility and injects the CO₂ into three regional geologic saline reservoirs for permanent storage.

The storage reservoirs comprise clastic Cretaceous aged formations that include the Lower Tuscaloosa Formation, the Washita-Fredericksburg interval and the Paluxy Formation, **Figure 2**. The storage reservoirs are capped by regionally extensive seals that include the marine Tuscaloosa shale, a shale at the top of the Washita-Fredericksburg interval, and a shale at the base of the Washita-Fredericksburg Group, **Figure 2**.

To assess the capacity of subsurface storage reservoirs and the sealing capability of their overlying cap rocks, several elements of geologic data were acquired and collected throughout the Kemper Regional Storage Complex. Geophysical well logs provide one of the best indirect methods to determine the quality of the subsurface geology for CO₂ storage. Importantly, geophysical logs can determine lithology, reservoir quality (i.e. porosity/permeability), water saturation, and reservoir thickness which are critical parameters to evaluate CO₂ storage potential.

To regionally characterize the geologic storage formations within the vicinity of the Kemper Regional Storage Complex, well logs from water wells and legacy oil wells that penetrate the storage geology were compiled, analyzed, and interpreted. In addition, three new wells (MPC 26-5 #1, MPC 34-1 #1, and MPC 10-4 #1) were drilled at the *ECO₂S* site in 2017 that penetrate the storage reservoirs in addition to their confining zones for subsurface characterization and injection monitoring, **Figure 1**. During drilling, core was collected from each well along the storage intervals to define the petrophysical properties of each reservoir. After core recovery, a suite of routine open hole geophysical logs was collected and used for additional geologic characterization.

This report provides a summary of all geophysical well logs that were used to characterize the storage reservoirs and associated seals within the vicinity of the Kemper Regional Storage Complex. Using a combination of subsurface logs from legacy oil wells, a water well, and the three new stratigraphic test wells, the storage site geology was assessed and characterized. Well log data from all sources were integrated into a subsurface geologic model which was used to determine CO₂ storage capacity, assess seal integrity, and model CO₂ injection plume behavior.

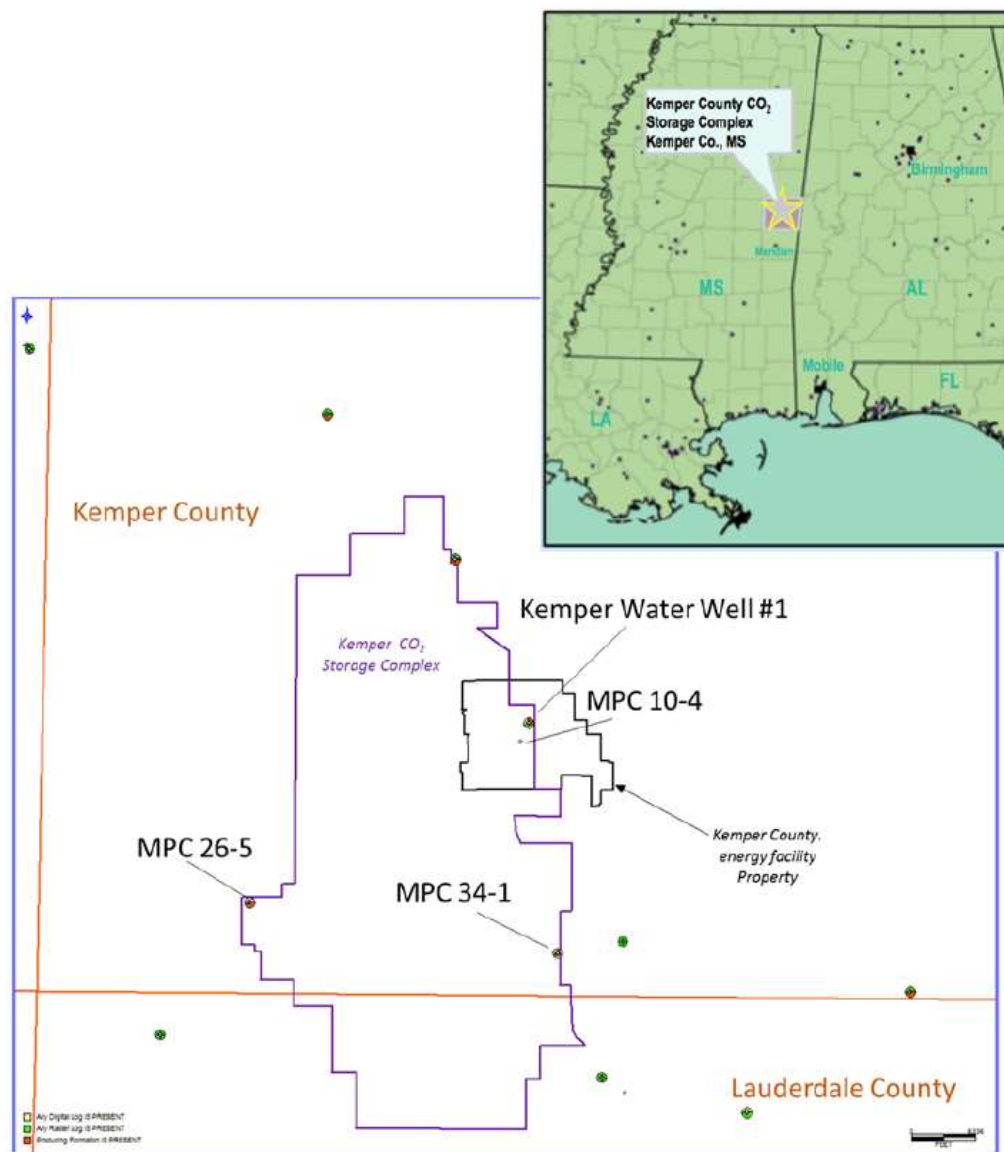


Figure 1. Map of the Kemper Storage Complex (blue) and Locations of the Three Geologic Characterization Wells.

Tertiary	Eocene	Lower Wilcox Group	Nanafolia Fm	Potential USDW
	Paleocene	Midway Group	Naheola Fm	Potential USDW
			Porter's Creek Clay	Regional Seal
Cretaceous	Upper	Selma Group	Chalk	Regional Seal
		Eutaw Formation		Potential USDW
		Tuscaloosa Group	Upper	Potential USDW
			Marine Shale	Regional Seal
			Lower & Massive Sand	★ Saline
	Lower	Washita Fredericksburg		★ Saline
		Paluxy Fm		★ Saline
Paleozoic Unconformity Ouachita Facies				

Figure 2. Stratigraphic Chart Showing the Saline Geologic Formations (green) with Excellent CO₂ Storage Capacity (yellow stars). Several Geologic Seals Overlay each Reservoir (pink). Geologic Formations Containing Potential USDW Aquifers are also shown (blue).

2.0 Routine Geophysical Well Logs

2.1 Overview

The regional geology of Kemper and Lauderdale County were assessed to evaluate the subsurface storage capacity using a combination of oil and gas wells, a water well, and three characterization/monitoring wells (*ECO₂S* CarbonSAFE wells). Kemper and Lauderdale Counties contain approximately 60 deep oil and gas wells and roughly 180 shallow public/private water wells. Within 10 miles of the CarbonSAFE test site, there are 13 deep well penetrations (legacy oil and gas wells), 1 deep water well (the Kemper County Energy Facility Water Test Well), and 3 recently drilled stratigraphic test wells (*ECO₂S* wells), **Figure 3**. Geophysical well logs from these wells were compiled, digitized, and loaded into Petra™ Geologic Modelling Software for interpretation.

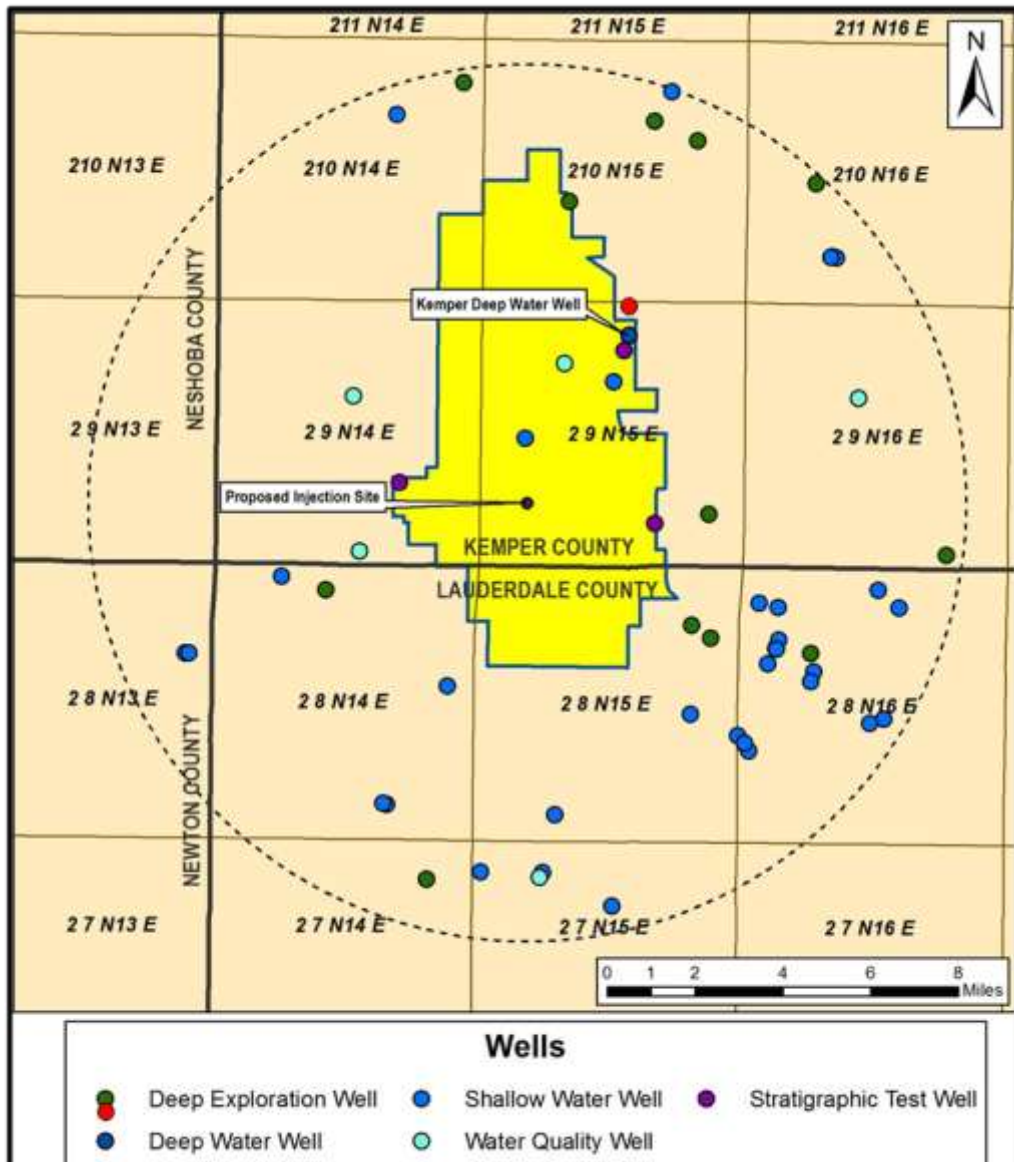


Figure 3. Location Map Showing the Position of the Kemper Storage Complex in Relation to a Selected Number of Oil/Gas Wells, Water Wells, and Stratigraphic Test Wells in Kemper County and Lauderdale County.

2.2 Legacy Oil and Gas Wells

Locations of legacy oil and gas wells in the vicinity of the Kemper Regional Storage Complex were determined using databases from Tomlin Geophysical Services (TGS), the Mississippi Oil and Gas Board (MSOGB), and the Alabama Geologic Survey. A map of oil and gas wells that contain either digital or raster geophysical well logs within a 10-mile radius of the Kemper Regional Storage Complex is shown in **Figure 4**. **Tables 1a, 1b, 1c, 1d, and 1e** indicate the available geophysical well logs that are associated with each oil and gas well within ~ 30 miles of the Kemper Regional Storage Complex.

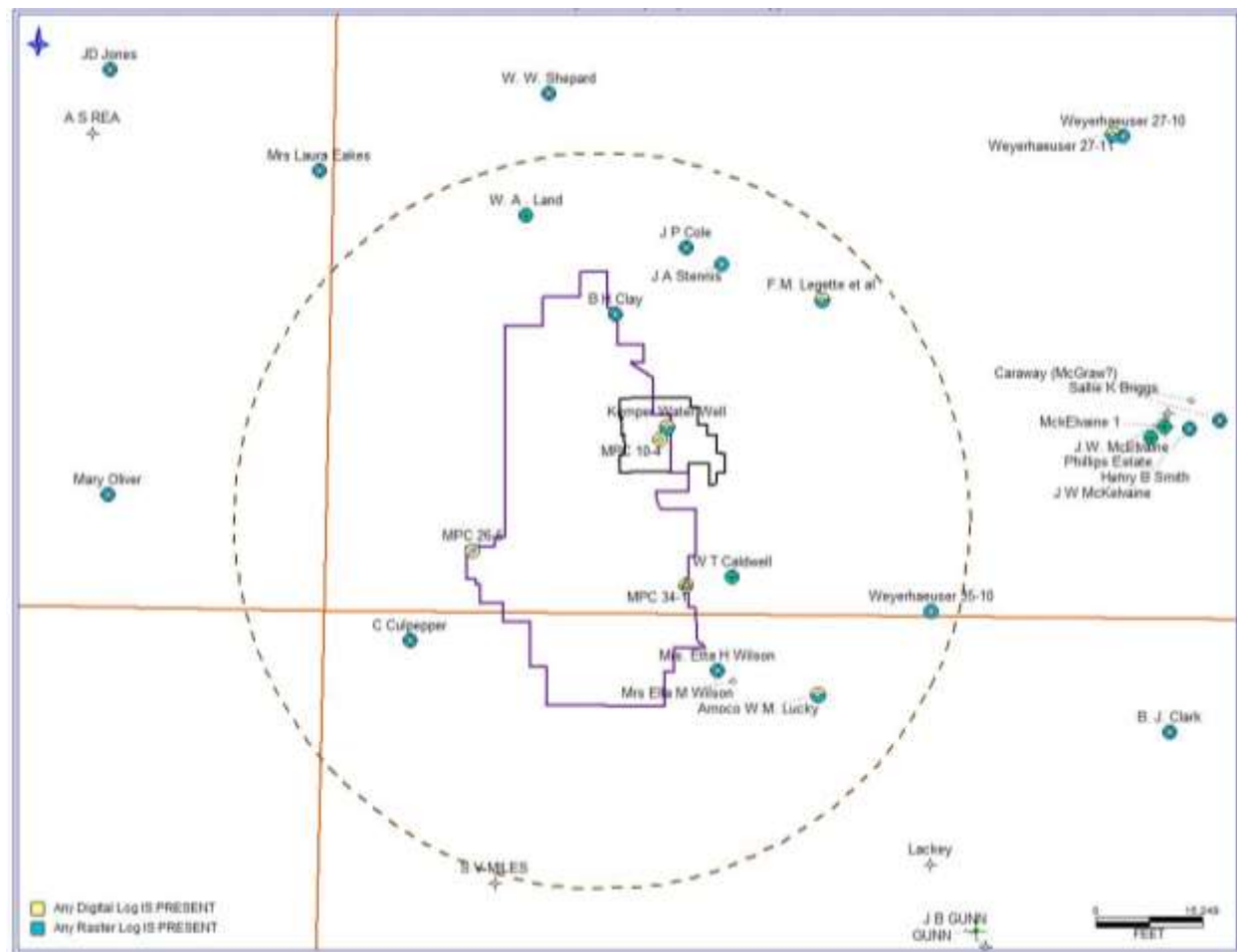


Figure 4. Locations of Legacy Oil and Gas Wells Near the Kemper Regional Storage Complex. The Dashed Line Indicates a 10-mile Radius Around the Center of the Injection Site.

Table 1a. List of Available Geophysical Well Logs from Legacy Oil and Gas Wells in Kemper County, MS.

Oil and Gas Well Logs in Kemper County						
Open Hole Logs	Sparkman 20-12 #1	Weyerhaeuser 27-6 #1	Weyerhaeuser 27-11 #1	Weyerhaeuser 26-10 #1	F.M. Legette et al 1	Amoco W.M. Lucky #1
Mud Log						
Caliper	x	x	x	x	x	x
Spontaneous Potential						
Spontaneous Potential from Resistivity		x	x	x	x	x
Gamma Ray						x
Gamma Ray from Resistivity	x	x	x	x		
Gamma Ray from Sonic					x	x
Sonic		x			x	x
Induction	x	x	x	x	x	
Normal Resistivity						
Shallow Focused Resistivity					x	
Laterolog Resistivity						x
Microlaterolog Resistivity						x
Neutron Porosity	x	x	x	x		x
Density Porosity		x	x	x		x
Bulk Density						
Structural Dip Meter						
Elemental Spectroscopy						
Formation Photoelectric Effect		x	x		x	

Table 1b. List of Available Geophysical Well Logs from Legacy Oil and Gas Wells in Kemper County, MS.

Oil and Gas Well Logs in Kemper County						
Open Hole Logs	Weyerhaeuser 27-11	H.W. Henry	W.T. Caldwell #2	Mrs. Etta H Wilson #1	C Culpepper #1	Weyerhaeuser 35-10
Mud Log						x
Caliper	x	x				x
Spontaneous Potential			x	x	x	x
Spontaneous Potential from Resistivity	x	x				
Gamma Ray						x
Gamma Ray from Resistivity	x					
Gamma Ray from Sonic						
Sonic		x	x	x	x	x
Induction	x	x				x
Normal Resistivity		x				x
Shallow Focused Resistivity						
Laterolog Resistivity						
Microlateralog Resistivity						x
Neutron Porosity	x					x
Density Porosity	x					x
Bulk Density						
Structural Dip Meter						x
Elemental Spectroscopy						x
Formation Photoelectric Effect						

Table 1c. List of Available Geophysical Well Logs from Legacy Oil and Gas Wells in Kemper County, MS.

Oil and Gas Well Logs in Kemper County						
Open Hole Logs	J A Stennis #1	J P Cole #1	B H Clay #1	W.A. Land #1	Weyerhaeuser 27-10 #1	Am Fleming Estate #1
Mud Log					x	x
Caliper					x	
Spontaneous Potential	x	x	x	x		x
Spontaneous Potential from Resistivity						
Gamma Ray					x	
Gamma Ray from Resistivity						
Gamma Ray from Sonic						
Sonic						
Induction		x	x	x		x
Normal Resistivity	x	x				x
Shallow Focused Resistivity		x				
Laterolog Resistivity						
Microlateralog Resistivity					x	
Neutron Porosity						
Density Porosity						
Bulk Density						
Structural Dip Meter					x	
Elemental Spectroscopy						
Formation Photoelectric Effect						

Table 1d. List of Available Geophysical Well Logs from Legacy Oil and Gas Wells in Kemper County, MS.

Oil and Gas Well Logs in Kemper County						
Open Hole Logs	Sallie K. Briggs #1	Henry B. Smith #1	J.W. McElvaine #1	Phillips Estate #1	Flintkote #2	B.J. Clark #1
Mud Log				X		
Caliper						
Spontaneous Potential	X	X	X	X		X
Spontaneous Potential from Resistivity					X	
Gamma Ray						
Gamma Ray from Resistivity						
Gamma Ray from Sonic					X	
Sonic						
Induction	X	X	X	X	X	X
Normal Resistivity		X		X		
Shallow Focused Resistivity						
Laterolog Resistivity						
Microlateralog Resistivity						
Neutron Porosity						
Density Porosity						
Bulk Density						
Structural Dip Meter						
Elemental Spectroscopy						
Formation Photoelectric Effect						

Table 1e. List of Available Geophysical Well Logs from Legacy Oil and Gas Wells in Kemper County, MS.

Oil and Gas Well Logs in Kemper County				
Open Hole Logs	Flintkote #1	Mrs Laura Eakes #1	Flinkote Forest #1	W.W. Shepard #1
Mud Log		x		
Caliper				
Spontaneous Potential	x	x	x	x
Spontaneous Potential from Resistivity				
Gamma Ray	x			
Gamma Ray from Resistivity				
Gamma Ray from Sonic				
Sonic	x			
Induction	x		x	x
Normal Resistivity		x	x	x
Shallow Focused Resistivity	x			
Laterolog Resistivity		x		
Microlaterolog Resistivity				
Neutron Porosity				
Density Porosity				
Bulk Density				
Structural Dip Meter				
Elemental Spectroscopy				
Formation Photoelectric Effect				

2.3 The Kemper Deep Water Test Well

A deep water test well (the Southern Company #1 well) was drilled by the Southern Company in October, 2007 immediately north of the present day MPC 10-4 test well. The well was drilled to a total depth of 3950 ft. and logged with Spontaneous Potential, Gamma Ray, and Induction Resistivity tools, **Figure 5**. The Kemper Water Well #1 only penetrates the uppermost sealing unit and two of the 3 potential storage reservoirs (the Marine Tuscaloosa shale, the Dantzler sand just below the Tuscaloosa 'Massive' sand, and the uppermost portion of the Washita-Fredericksburg) with the remaining storage and confining units occurring below at greater depth. The well was permanently plugged and abandoned in 2019.

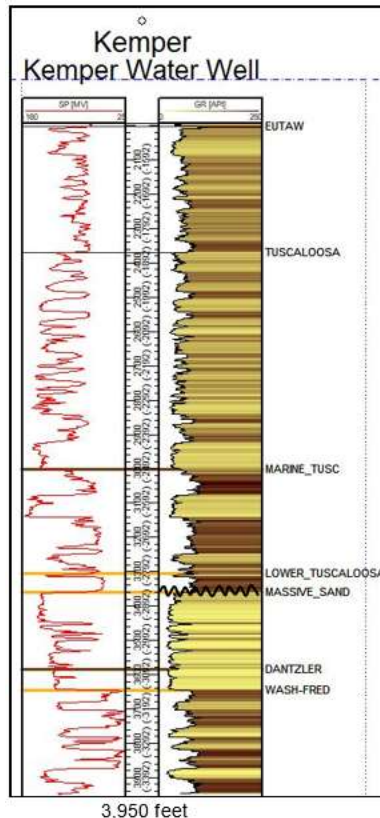


Figure 5. Geophysical Well Log Depicting the Storage and Confining Units Penetrated by the Kemper Deep Water Test Well (2007). The left track (red curve) indicates spontaneous potential measurements. The right track indicates the gamma ray logging response. Light colors indicate clean sand while dark colors indicate shale. The water well also contains induction resistivity logs but they are not pictured here.

2.4 New well logs (triple combo, mud logs)

The three new geologic characterization/monitoring wells (MPC 26-5 #1, MPC 34-1 #1, and MPC 1-4 #1) were drilled in close proximity to the Kemper County Energy facility, **Figure 6**. Each characterization well penetrated the storage reservoir and confining units. After whole and sidewall cores were recovered at each well, geophysical well logs were run in open hole at different depths, **Table 2**. Mud logs were acquired during drilling in addition to a suite of 'Triple Combo' geophysical logs, **Figure 7, 8, 9**.

Mud logging was used to create a detailed lithological column of the borehole by recording the rock cuttings brought to the surface by drilling mud. Rock cuttings are also evaluated for hydrocarbon shows. In addition, mud logging procedures also record gas shows if present. Mud logs were used in combination with geophysical well logs to define formation tops, pick core points, evaluate potential hydrocarbon zones, and assess geologic risks (sour gas, fault zones, drilling hazards). Representative cuttings were

collected and recorded every 20 ft. during mud logging operations. Mud logs were gathered from the surface of each well to total depth (TD). Mud logs did not indicate the presence of any hydrocarbon zones in the ECO_2S characterization wells.

The 'Triple Combo' logs contain Array Induction (Resistivity), Gamma Ray (GR), Spontaneous Potential (SP), Sonic, Differential Caliper, Formation Photoelectric Factor, Density Porosity, Neutron Porosity, and Borehole Profile log measurements.

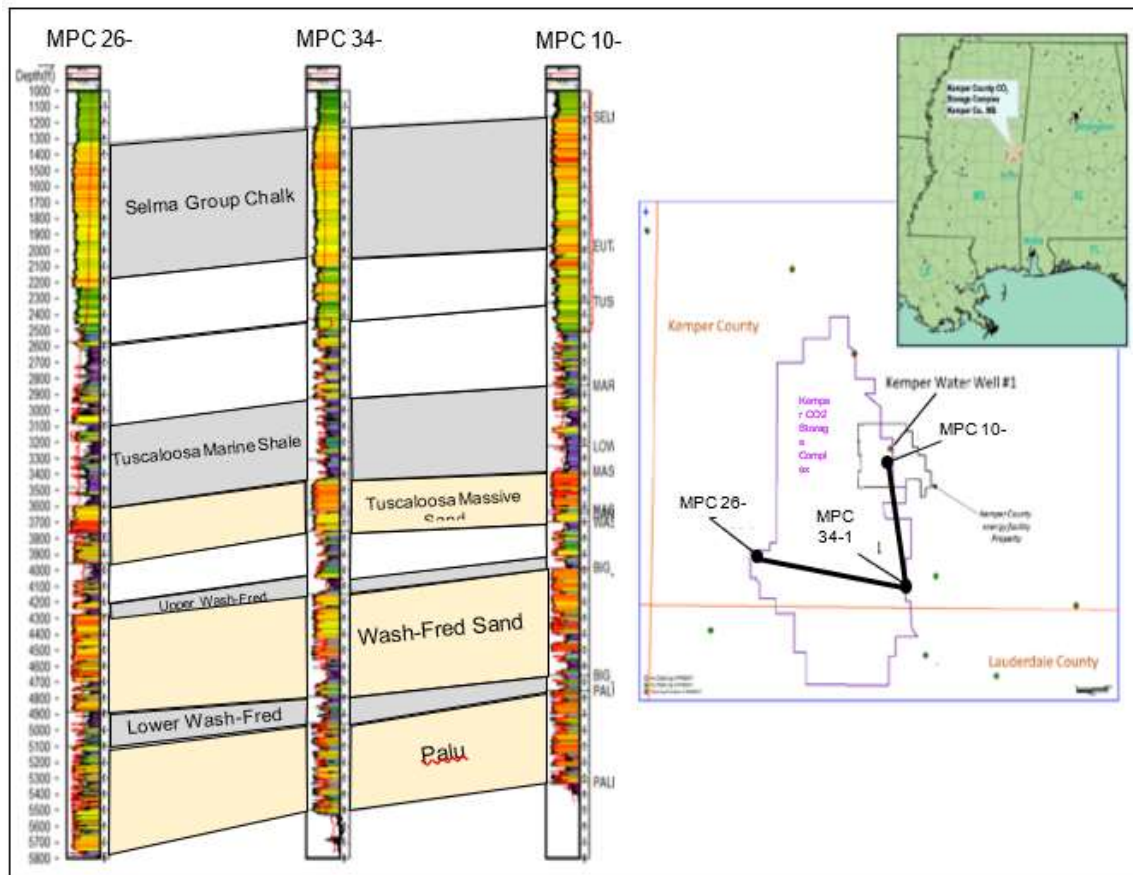


Figure 6. (Left) Cross-Section Showing the Three Newly Drilled ECO_2S CarbonSAFE Wells: MPC 26-5, 34-1, and 10-4. The tops of each geologic formation are correlated between each well. (Right) Location map of the ECO_2S Storage Complex showing the location of the wells.

Table 2. A List of Geophysical Well Logs for Each of the Characterization Wells at the Kemper Storage Complex.

Well Logs						
Logs	MPC 26-5		MPC 34-1		MPC 10-4	
Surface Casing						
Borehole Profile	x	100 - 2,489 ft.	x	90 - 506 ft.	x	103 - 2,008 ft.
Platform Express: Compensated Neutron, Lithodensity, Array Induction, GR, SP, Caliper	x	100 - 707 ft.	x	91 - 506 ft.	x	104 - 2,008 ft.
Cement Bond Log	x	0 - 2,434 ft.	x	surface - 2,334 ft.	x	surface - 2,470 ft.
Temperature	x	0 - 2,434 ft.	x	surface - 2,334 ft.		surface - 2,330 ft.
Long String						
Mudlog	x	2,500 - 5,877 ft.	x	2,500 - 5,750 ft.	x	2,520 - 5,440 ft.
Borehole Profile	x	2,491 - 5,869 ft.	x	2,495 - 5,740 ft.	x	2,520 - 5,440 ft.
Combinable Magnetic Resonance Tool	x	3,000 - 5,855 ft.	x	2,495 - 5,740 ft.	x	2,520 - 5,440 ft.
Platform Express: Compensated Neutron, Lithodensity, Array Induction, GR, SP, Caliper	x	2,491 - 5,869 ft.	x	2,495 - 5,740 ft.	x	2,520 - 5,440 ft.
Sonic Scanner	x	2,491 - 5,877 ft.	x	2,830 - 5,748 ft.	x	2,520 - 5,440 ft.
Cement Bond Log	x	100 - 5,740 ft.	x	400 - 5,600 ft.	x	100 - 5,330 ft.
Formation Micro Imager	x	2,491 - 5,877 ft.	x	2,830 - 5,748 ft.	x	2,600 - 5,440 ft.
Mechanical Sidewall Coring Tool	x	2,360 - 5,715 ft.				
Additional Logs						
Pulsed Neutron Capture	x	2,475 - 5863 ft.	x	2,485 - 5,500 ft.	x	2,016 - 5,440 ft.

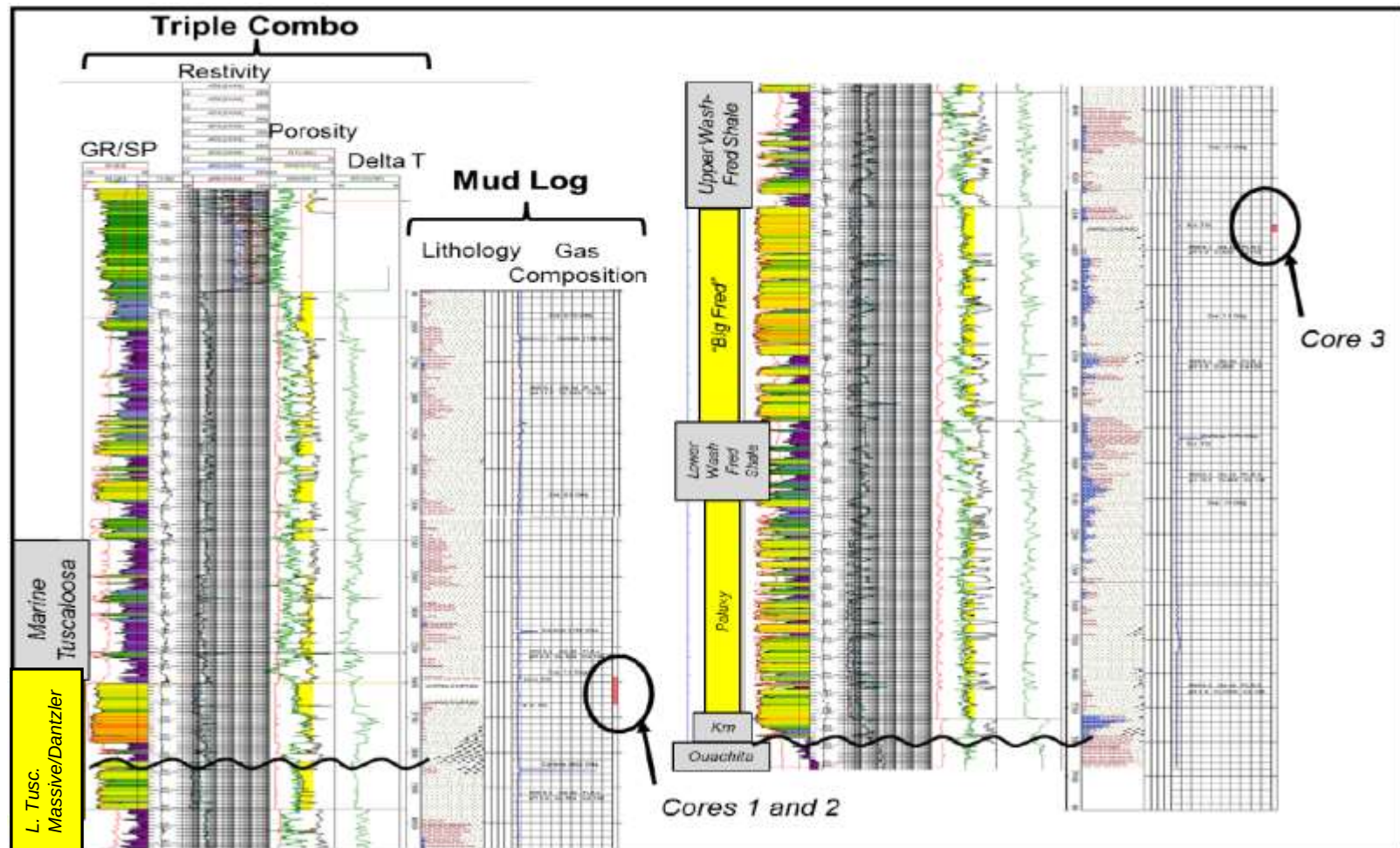


Figure 7. MPC 26-5 #1. Composite Image of the Triple Combo Open Hole Log and Mud Log. Core points are identified to the right of the log tracks. The first track contains a color filled natural gamma ray curve. Warmer colors (yellow) indicate reservoir intervals (sandstones). Cooler colors (green/blue) indicate confining units (mudstones). The third track shows Neutron and Density porosity curves. The Density Porosity Curve is filled (yellow) when Density Porosity is > 20%.

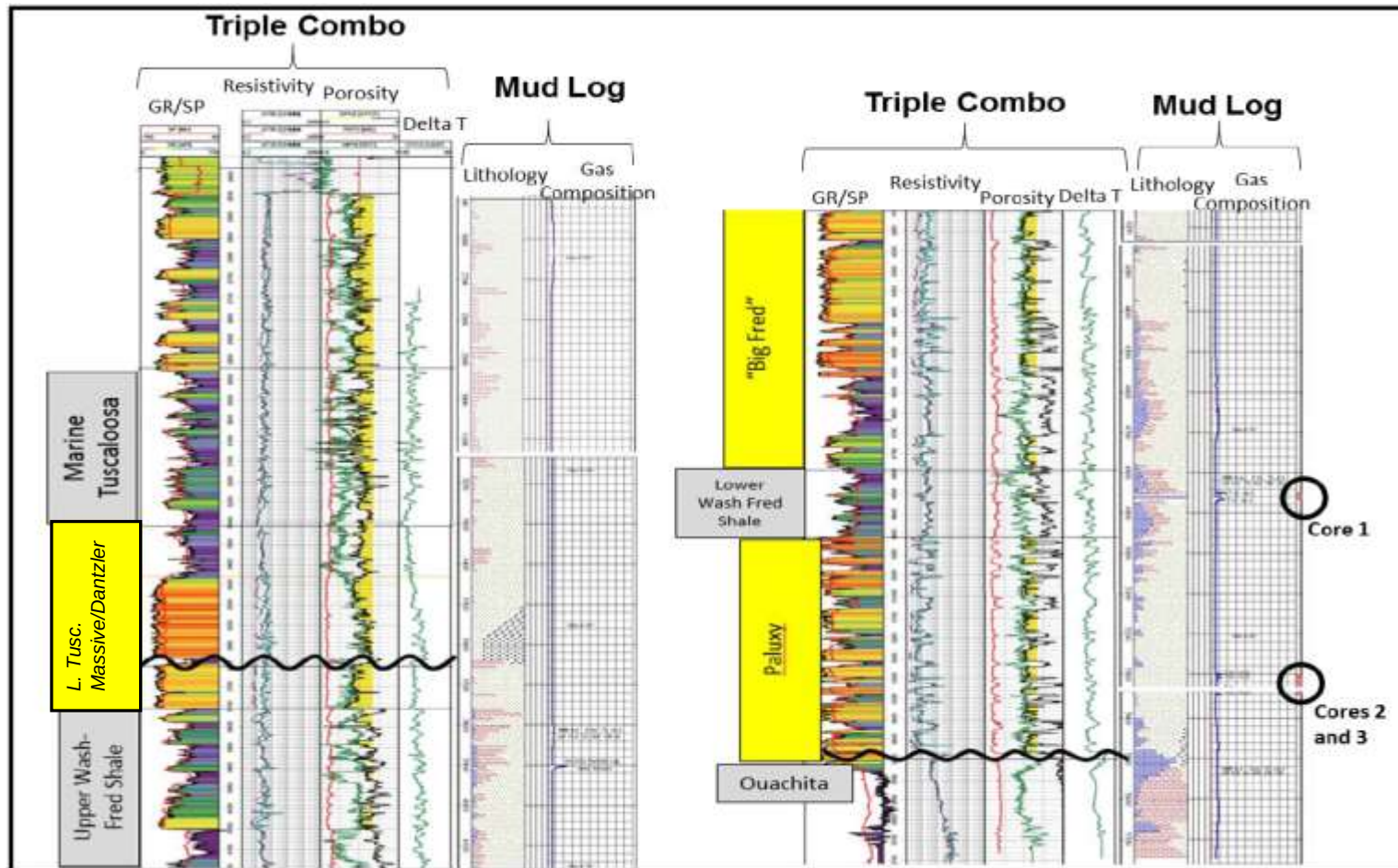


Figure 8. Well MPC 34-1 #1. Composite Image of the Triple Combo Open Hole Log and Mud Log. Core points are identified to the right of the log tracks. The first track contains a color filled natural gamma ray curve. Warmer colors (yellow) indicate reservoir intervals (sandstones). Cooler colors (green/blue) indicate confining units (mudstones). The third track shows Neutron and Density porosity curves. The Density Porosity Curve is filled (yellow) when Density Porosity is > 20%.

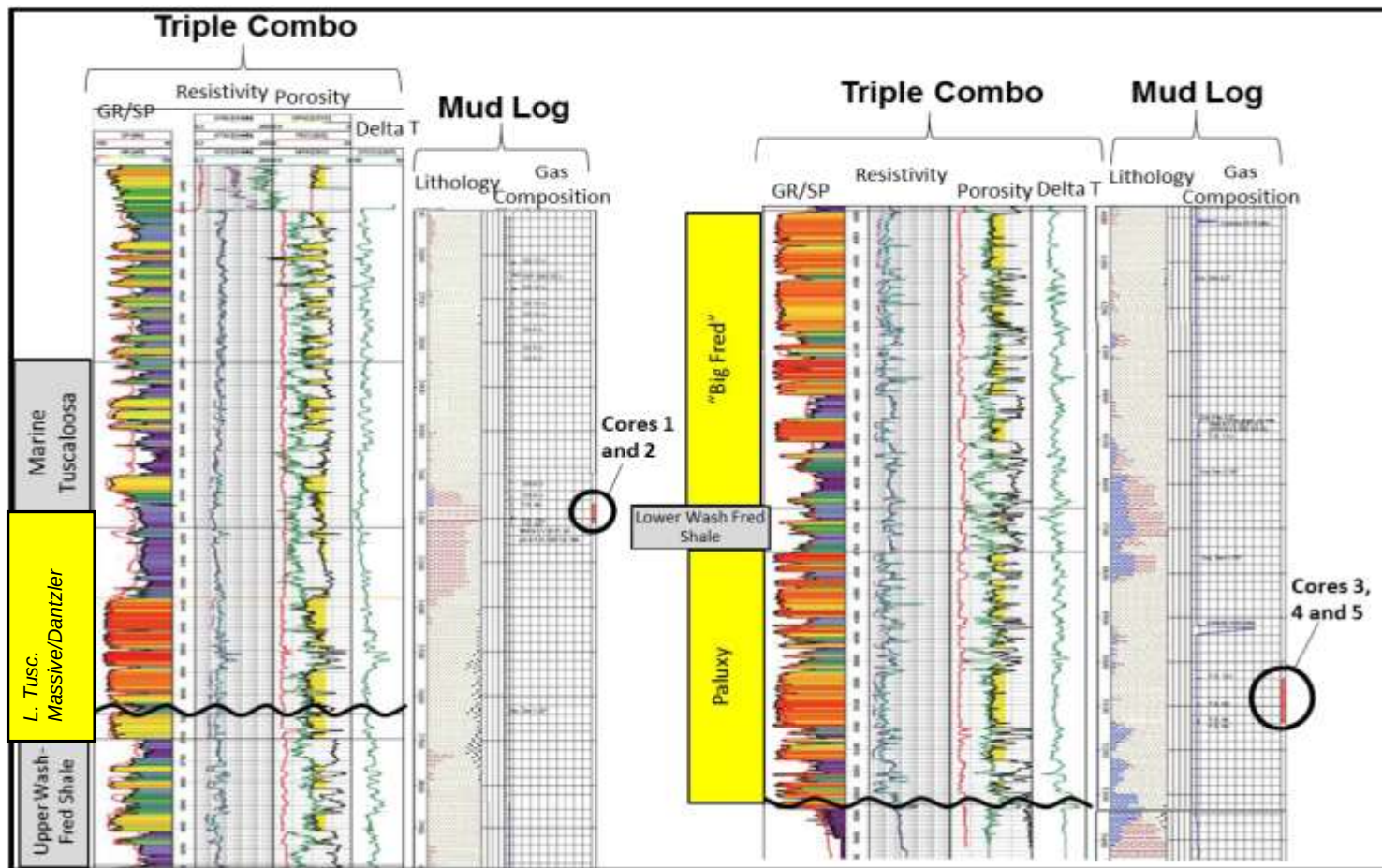


Figure 9. Well MPC 10-4 #1. Composite Image of the Triple Combo Open Hole Log and Mud Log. Core points are identified to the right of the log tracks. The first track contains a color filled natural gamma ray curve. Warmer colors (yellow) indicate reservoir intervals (sandstones). Cooler colors (green/blue) indicate confining units (mudstones). The third track shows Neutron and Density porosity curves. The Density Porosity Curve is filled (yellow) when Density Porosity is > 20%.

3.0 Non-routine Processed Geophysical Logs

Non-routine processed geophysical logs were acquired at the three characterization wells (MPC 26-5 #1, MPC 34-1 #1, and MPC 1-4 #1) by Schlumberger and include Combinable Magnetic Resonance (CMR), Dipole Sonic, and Formation Micro Imager (FMI) tools that measure the properties of the storage formation geology in open-hole. In addition, Pulsed Neutron Capture (PNC) logs were acquired after the wells were cased and cemented. A list of non-routine processed geophysical logs and depths of investigation at each *ECO₂S* characterization well are presented in **Table 2**. The following subsections provide a basic summary of how each non-routine geophysical tool functions and the basic rock/fluid properties that were measured for this project.

3.1 CMR Logs

The CMR Tool is Schlumberger's nuclear magnetic resonance (NMR) tool. NMR tools use a strong magnetic field to investigate the amount of hydrogen within the target formation to assess reservoir quality. When used with conventional logs, NMR logs provide a powerful method to evaluate porosity, pore distribution, and bound-/free-water volume without being influenced by formation lithology. NMR derived porosities are advantageous over density porosity, neutron porosity, and sonic porosity measurements which are heavily influenced by the presence of clay bound water and high gas saturations.

The CMR logs were used to establish the pore size distribution and pore throat sizes of the subsurface storage reservoirs within the Storage Complex. Using this approach, estimates of porosity and permeability with depth were determined from the CMR logs. The CMR derived porosity and permeability values were calibrated to porosity and permeability determined from recovered whole core and thus enable more accurate predictions of reservoir quality from well logs. Overall, the CMR log enables accurate characterization of storage geology reservoir quality when combined with core and the triple-combo well logs.

3.2 Dipole Sonic/Geomechanical Logs

The Dipole Sonic scanner provides important information regarding a wide range of petrophysical and geomechanical properties by measuring interval transit time (ΔT), compressional, and ΔT shear wave velocities of the lithologies within the borehole. For the *ECO₂S Project*, measured compressional and shear wave acoustic velocities from the Dipole Sonic logs were used to determine dynamic elastic constants of the reservoirs and confining units at each *ECO₂S* well. These geomechanical constants were used to estimate static elastic properties, in-situ horizontal stress, and rock strength. Chiefly,

these parameters gauge the rock strength of the reservoirs and confining units which can be evaluated during periods of elevated pore pressure due to CO₂ injection (see *Deliverable 6.4 - Thermo-Hydro-Mechanical Modeling Report*). In addition, these values were also used to assess the likelihood of induced seismicity related to CO₂ injection at the *Project ECO₂S* site.

3.3 FMI Log Interpretation

The FMI includes measurements of micro-resistivity, a dip meter, and a multi-arm caliper. The micro-resistivity tool measures the electrical resistivity of the formation and is logged at a very fine scale (< than one inch). Measurements of resistivity are used to determine the type of fluids within the pore space (i.e. hydrocarbons, fresh water, saline water) and the degree of water saturation. Also included in the tool is a four-arm caliper that measures borehole diameter. The dip meter can assess the orientation of structural and sedimentological features along the borehole.

The results of the FMI logs from the *Project ECO₂S* wells indicate consistently low resistivity contrast throughout the Kemper storage geology. These results indicate that either significant drilling mud invasion has occurred into the formation during drilling or that the storage reservoirs are completely saturated with saline brines. However, the FMI results provided enough data for researchers at Oklahoma State University to determine directional control on cross-bed and ripple orientation in the Paluxy sandstone (see *Deliverable 6.2.b – Geologic Framework for the Kemper Regional Storage Complex*).

3.4 PNC Logs

PNC logs are primarily used to monitor a reservoir by detailing the reservoir saturation response to injection operations. Pulsed neutron logging (PNL) tools are capable of making several different measurements sensitive to CO₂ in the formation and in the casing-formation annulus in cased wells. Therefore, the PNL can be used to monitor the formation fluids as well as identify mechanical integrity problems that may allow the CO₂ to migrate up the casing annuli. PNC logs were collected in May of 2019 on the MPC 10-4 wells and in May of 2020 on wells MPC 26-5 and MPC 34-1. On well MPC 34-1, tubing was pulled from the well prior to logging operations.

4.0 Reservoir Quality Interpretations from Geophysical Well Logs

4.1 Average Porosity Calculations from the 'Triple Combo' Well Logs

The 'Triple Combo' well logs (**Figures 7, 8, 9**) collected at the Phase II *ECO₂S* characterization wells were used to determine formation porosity with the neutron porosity (NPHI) and density porosity (DPHI) logging tools. These tools provide first order estimates of lithological porosity but are influenced by lithology, gas saturation, and fluid saturations. The density logging tool effectively measures the density of electrons within the target formation which corresponds to bulk density. Rocks containing ample pore space contain lower electron densities that result in low density porosity measurements. In contrast, the neutron tool quantifies the hydrogen concentration of shale-free formations by measuring the liquid filled porosity. Combining both the density porosity and neutron porosity logging responses is an industry standard method to estimate porosity.

For Project *ECO₂S*, the density porosity and neutron porosity measurements from the 'Triple Combo' logs collected at each characterization well were analyzed across the reservoir and confining intervals. Lithological sand reservoirs were primarily identified by evaluating intervals with low gamma ray and spontaneous potential log responses which indicate relatively clean (shale free) sands. Cleans sands with correspondingly high density and neutron porosity log responses suggest high porosity sand reservoirs. To estimate the porosity of each sand, the density porosity and neutron porosity values were averaged over the total depth of each net sand reservoir interval. Chiefly, average porosity for each interval combined with net thickness and gross rock volume were used to calculate total storage capacity estimates of each CO₂ storage reservoir at the Kemper Regional Storage Complex. Net storage reservoir thicknesses and total porosity determined from 'Triple Combo' log results are shown in **Table 3**.

Table 3. Total Porosity from Triple Combo Logs Run in the Three Kemper Characterization Wells

	MPC 26-5		MPC 34-1		MPC 10-4	
Formation	Net Sand (ft.)	PHIA (%)	Net Sand (ft.)	PHIA (%)	Net Sand (ft.)	PHIA (%)
Massive Sand	167	30	198	30	181	31
Dantzler	127	33	94	30	86	32
Big Fred	394	28	348	28	378	27
Paluxy	445	28	363	27	406	28

4.2 Core and CMR Derived Permeability

Core was successfully recovered from wells MPC 34-1 and MPC 10-4 while core from well MPC 26-5 was too damaged for analysis upon recovery. Routine core analysis (RCA) derived porosity and permeability measurements were determined for whole cores recovered from wells MPC 34-1 and MPC 10-4. For routine core analysis two methods are used to calculate gas permeability to air: steady state and unsteady state (Klinkenberg). For steady-state permeability to air, a clean, dry sample is placed in a core holder and air is injected at constant pressure until the gas production rate and flow rate stabilize at the opposite end of the sample. The pressure differential between the two ends of the core and the flow rate are recorded. These measurements are input to the integrated form of Darcy's law for a compressible fluid to calculate permeability. Unsteady-state permeability involves the transient analysis of a pressure pulse decay in a gas saturated sample. Gas slippage is determined as a function of the gas pressure decay. Gas used for unsteady-state permeability is generally helium. Unsteady state permeability is considered to be more accurate because the method relies on the measurement of pressure versus time instead of pressure versus flow rate. Klinkenberg permeability is more representative of permeability under reservoir conditions than steady-state permeability to air. For this reason, the unsteady state (Klinkenberg) results are used for the petrophysical analysis. Core data and special core analyses provide the relationship between permeability and porosity; log data makes it possible to identify and correlate individual permeable flow units. Full core analysis results can be found in *Deliverable 6.1 – Core Analysis Report*.

The CMR logs were used to extrapolate permeability throughout the storage geology. An example of the CMR log for well MPC 10-4 is shown in **Figure 10**. CMR uses a strong magnetic field to polarize the spin axes of protons (i.e. hydrogen) in a specific direction. In addition, an oscillatory magnetic field is pulsed so that the formation protons are in resonance which produces a strong signal, or echo, that decays over time. The exponential decay of this signal over time is defined as the 'relaxation time' which is related to the number of decaying echoes produced by each fluid in the formation. Thus, the relaxation time of a given fluid within a pore is proportional to the size of the pore. Water within smaller pores generate shorter relaxation times while larger pores produce longer relaxation times. The distribution these of relaxation times indicate the pore size distribution of the formation.

The measured relaxation times indicate the pore size distribution which can be used to estimate the permeability of the target formation. As the pore size distribution increases, pore space becomes more interconnected and results in increasingly higher permeability. For the *ECO₂S* Project, permeability was derived from CMR logs using the

Timur-Coates model that uses the relative proportion of free-fluid to capillary-bound fluid within the effective porosity to estimate permeability following,

$$K_{TIM} = a_2(10^4) \left(\frac{\Phi_{FF}}{\Phi_{BF}} \right)^{b_2} (\Phi_{CMR})^{c_2}$$

Where the CMR derived permeability (K_{TIM}) is a product of a constant (a_2), the free- and bound- fluid porosity ratio exponent ($\left(\frac{\Phi_{FF}}{\Phi_{BF}} \right)^{b_2}$), and the porosity exponent ($\Phi_{CMR}^{c_2}$). This method produces a continuous permeability curve over the log interval that is shown in **Figure 10**.

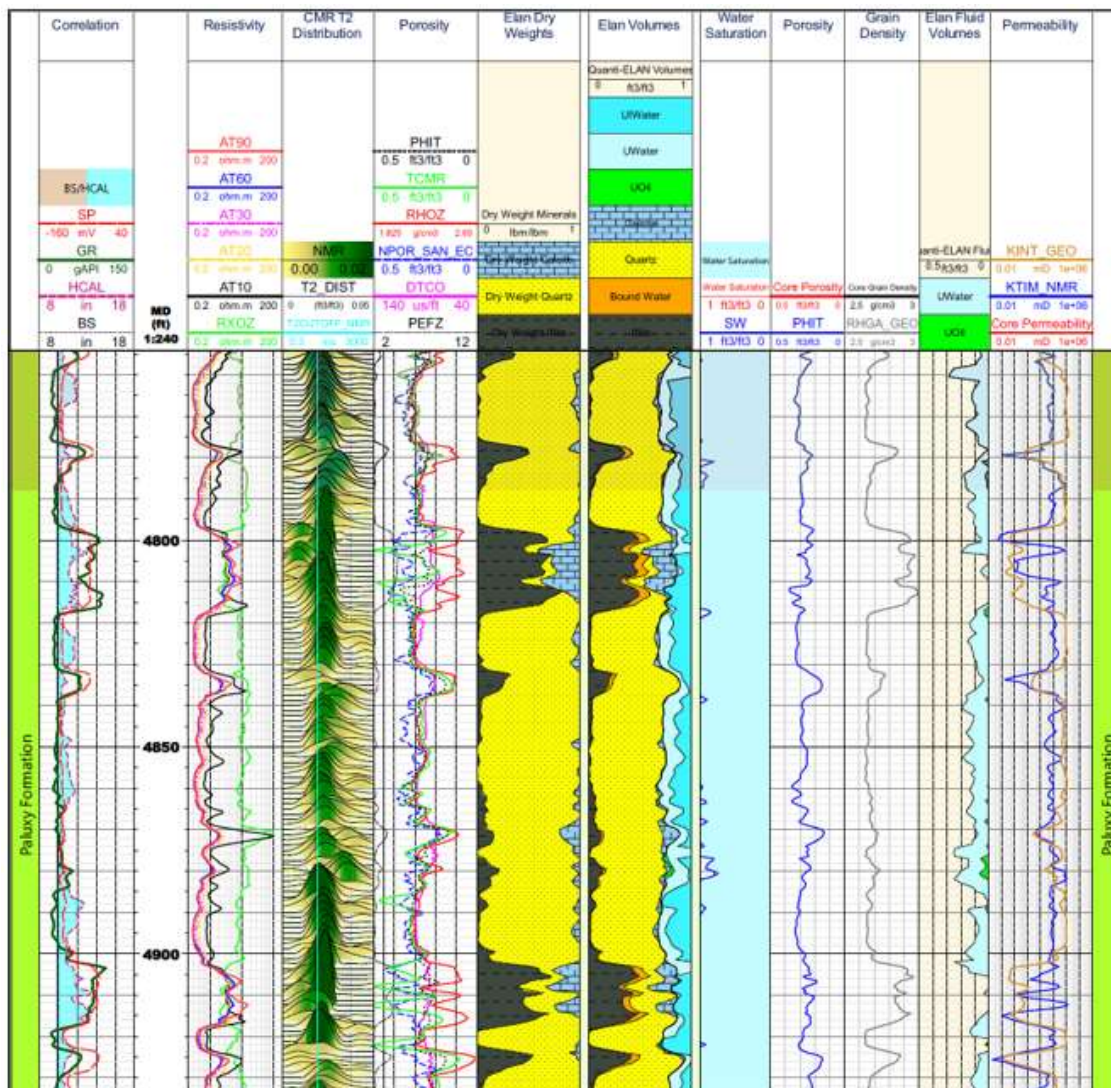


Figure 10. Schlumberger CMR Well Log for Well MPC 10-4 at Kemper County, Mississippi Through the Uppermost Paluxy Formation. The CMR log occurs in the third track from the left and is indicated by 'NMR'. An extrapolated CMR derived permeability is shown on the farthest right track in blue (KTIM_NMR).

5.0 Stratigraphic Cross-Sections

After compiling a database of geophysical well logs for Kemper County, Mississippi in Petra™ geologic modelling software, two cross-sections were drawn through the *ECO*₂S storage complex to assess the regional continuity of the storage geology. These cross-sections were drawn through the *ECO*₂S wells in addition to nearby legacy oil wells with available logs that penetrated the storage geologic formations. Both north-south and west-east cross-sections were drawn through the immediate *ECO*₂S project site (**Figure 11**) and are shown in **Figures 12 and 13**. The tops of each storage formation, including sand reservoirs and confining shale units were interpreted in the geophysical well logs and correlated across each well to characterize the lithological continuity of each unit.

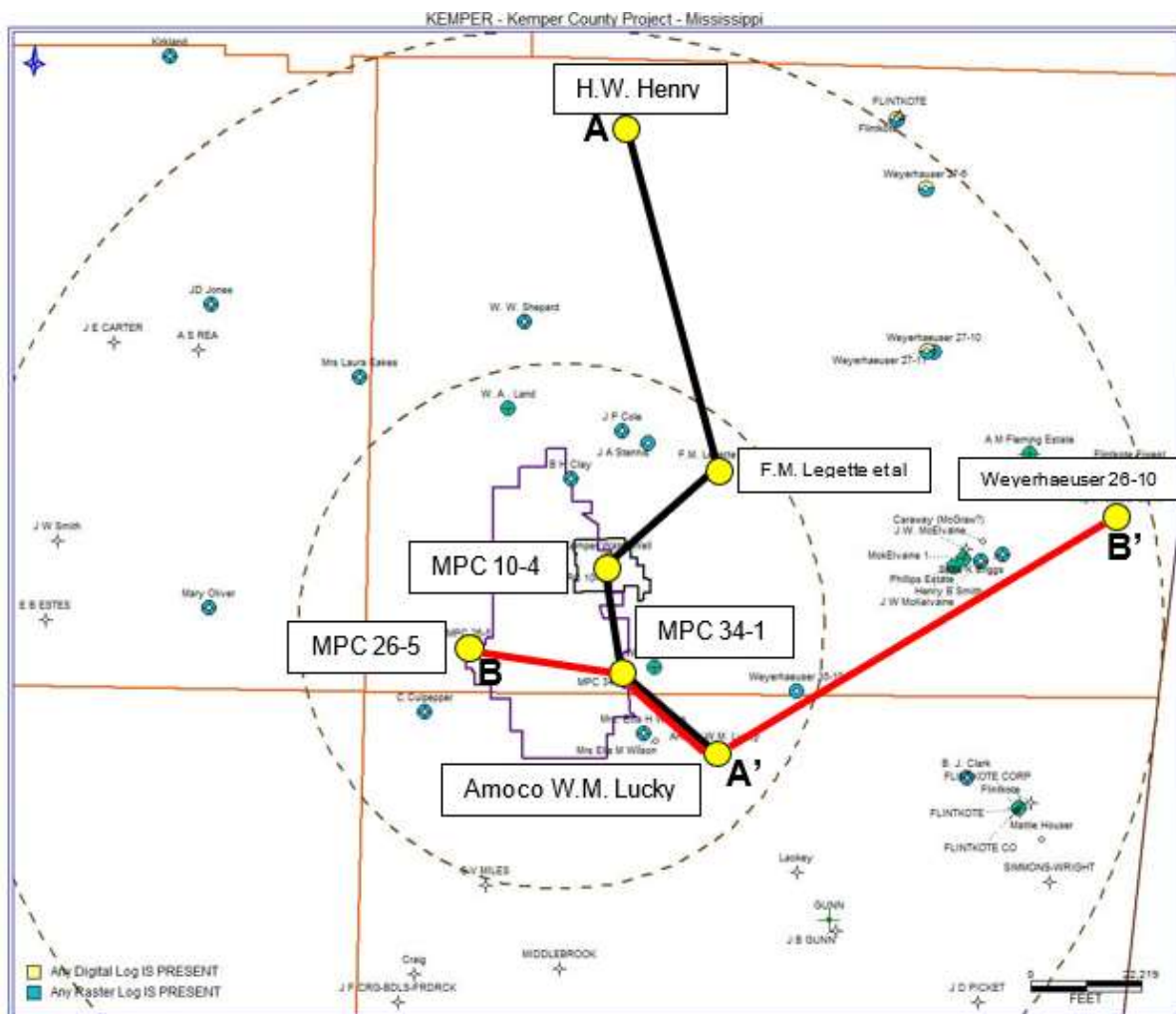


Figure 11. Location Map of the *ECO*₂S Project Site and the Location of the Characterization Wells and Legacy Oil Wells Within the Vicinity of the Storage Complex.

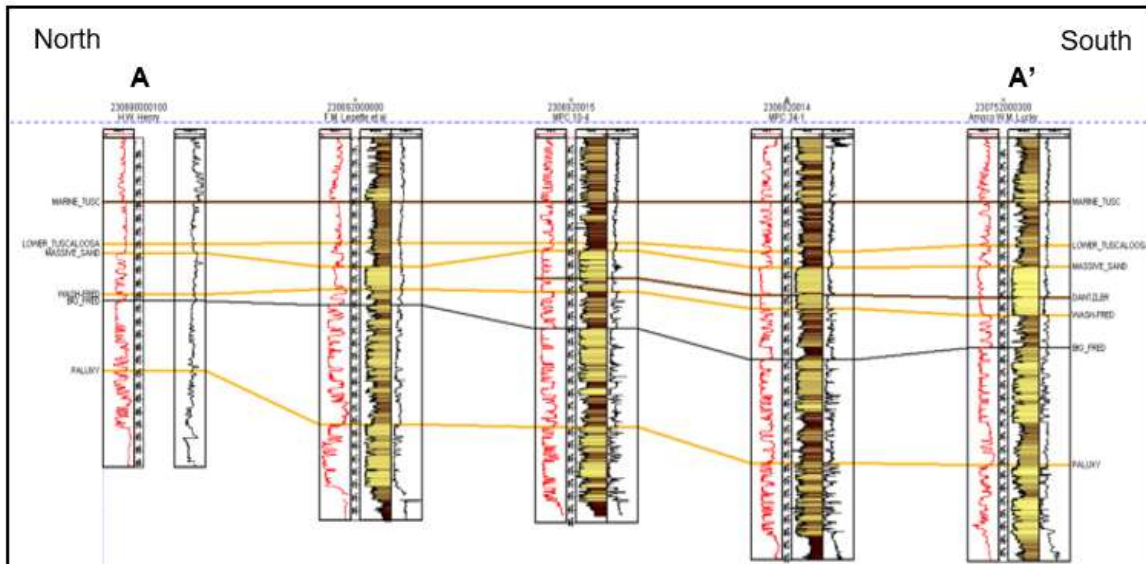


Figure 12. North-South Cross-Section Through the ECO₂S Storage Complex Through Both Legacy and Characterization Wells. The cross-section is flattened on the Marine Tuscaloosa stratigraphic horizon. The log tracks from left to right indicate Spontaneous Potential, Gamma Ray, and Deep Resistivity. Well H.W. Henry does not have an available Gamma Ray log. The tops of pertinent storage geology are correlated between each well.

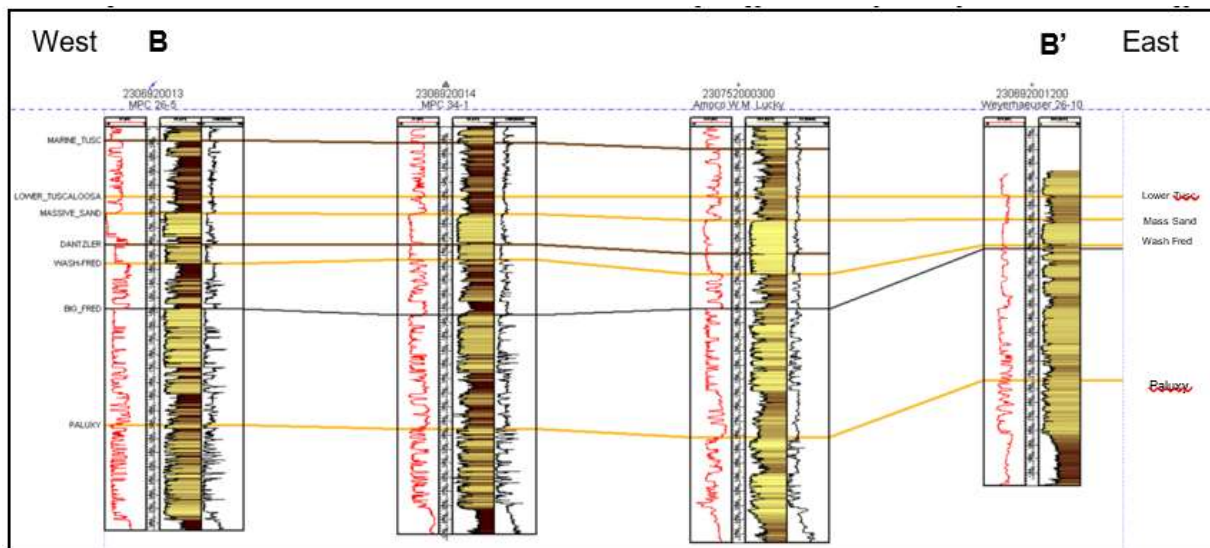


Figure 13. West-East Cross-Section Through the ECO₂S Storage Complex Through Both Legacy and Characterization Wells. The cross-section is flattened on the Lower Tuscaloosa stratigraphic horizon. The log tracks from left to right indicate Spontaneous Potential, Gamma Ray, and Deep Resistivity. Well Weyerhaeuser 26-10 does not have an interpreted top for the Marine Tuscaloosa shale.

6.0 Conclusions

Numerous geophysical well logs were compiled, gathered, and analyzed for Phase II of the *ECO₂S* CarbonSAFE initiative to assess deep underground CO₂ storage in Kemper County, Mississippi. A combination of routine geophysical logs and non-routine processed geophysical logs were compiled and gathered to interpret and assess the storage capability of several stacked subsurface reservoirs and confining units. Data from the logs were incorporated into a geologic model of the storage site and used to estimate important reservoir quality parameters such as porosity and permeability.

Routine geophysical logs that were compiled include data from legacy oil/gas wells and the Kemper Water Well. In addition, new well logs at the three newly drilled *ECO₂S* wells were also used to characterize the storage geology using both Triple Combo and mud logs. The Triple Combo logs contain density porosity and neutron porosity logs which were used to calculate average porosity across the storage reservoir sand intervals.

The non-routine processed geophysical logs comprise CMR, Dipole Sonic, and FMI logs and were used to provide more detail on the petrophysical properties of the storage geology. The Dipole Sonic logs primarily assessed the geomechanical/rock strength properties of the storage geology. The FMI logs evaluated a largely unchanging low micro-resistivity response from lithologies within the borehole. Chiefly, the CMR logs were used to estimate permeability across the entire logged intervals of the wells within the *ECO₂S* Storage Complex. Estimates of both porosity and permeability was used to determine the CO₂ capacity of the entire storage complex.

Lastly, two cross-sections were drawn through the storage complex using a combination of both legacy oil/gas wells and the *ECO₂S* characterization wells. One cross-section was drawn from north to south while a second cross-section was drawn from west to east. The geology was interpreted throughout the storage site and pertinent geological horizons were interpreted from wells. Overall, this provided important information on the continuity and thickness of the storage geology.